

L8: Community Size Distribution

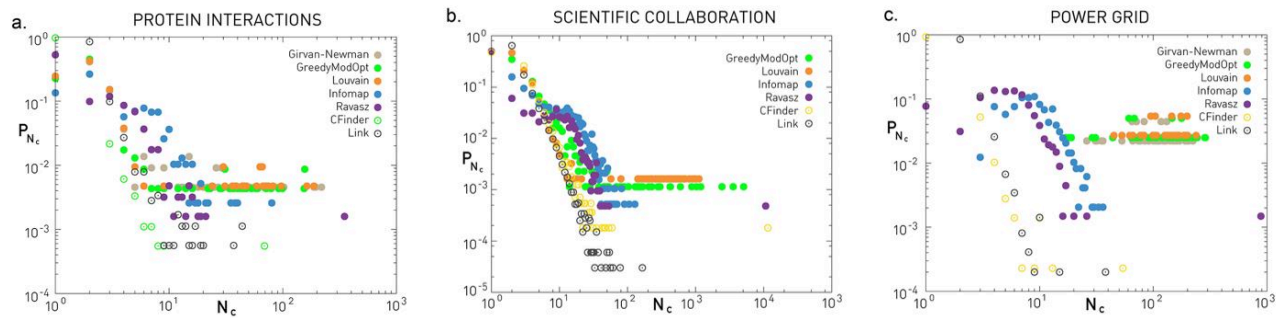


Figure 9.29 from *Network Science* by Albert-László Barabási

The GN benchmark creates communities of identical size. How realistic is this assumption, however?

Even though we typically do not have the ground-truth, there is increasing evidence that most networks have a power-law community size distribution.

This implies that most communities are quite small, maybe a tiny percentage of all nodes, but there are also few large communities with comparable size to the whole network.

The three plots shown here refer to three different networks (*the same datasets we have seen many times in the past – discussed in your textbook*).

The plots show the probability distribution of the discovered community sizes according to six different algorithms -- we have discussed all of them except Infomap.

Even though there are major differences between the six algorithms (*we will return to this point later in this lesson*), they all tend to agree about the Protein Interaction network and the Scientific Collaboration network that the size distribution is heavy-tailed and it can be approximated by a power-law.

The situation is less clear with the Power Grid network – there are major differences between the six algorithms in terms of the shape of the community size distribution.

Food for Thought

The previous plots show several points with the same y-axis value at the right tail of the distribution. What does this mean about the number of communities with that size?